

# Project Design Phase – II

## Data Flow Diagram & User Stories

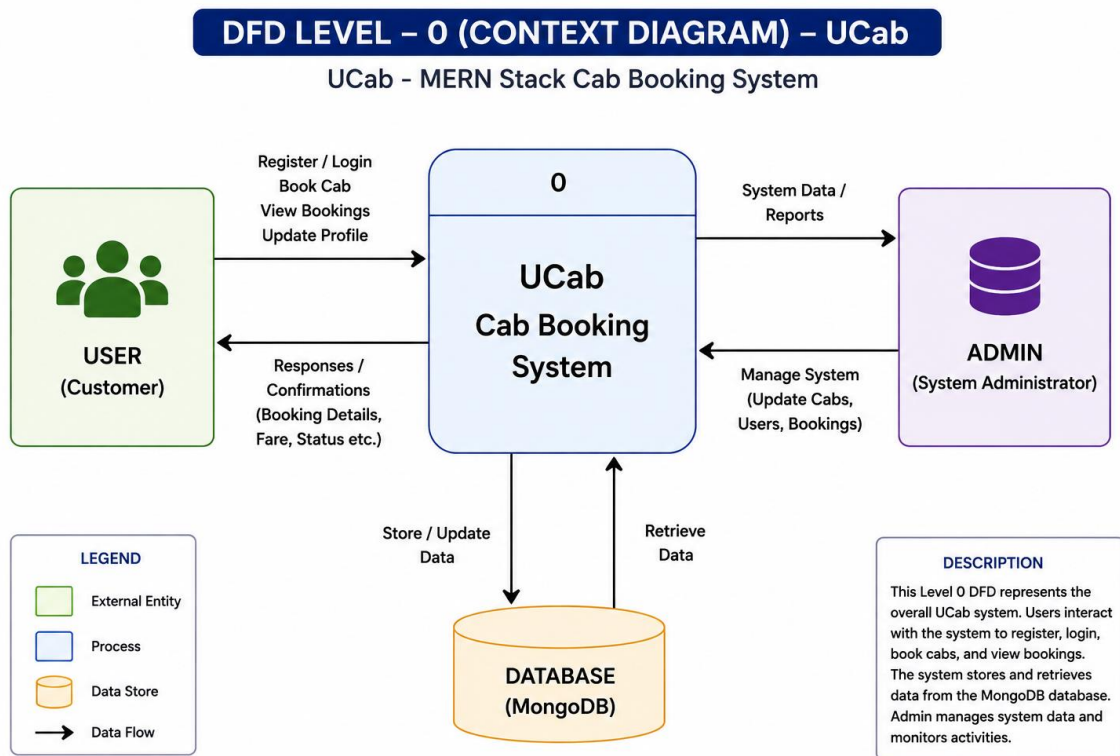
**Date:** 10-07-2026

**Team ID:** SWTID-2026-5429

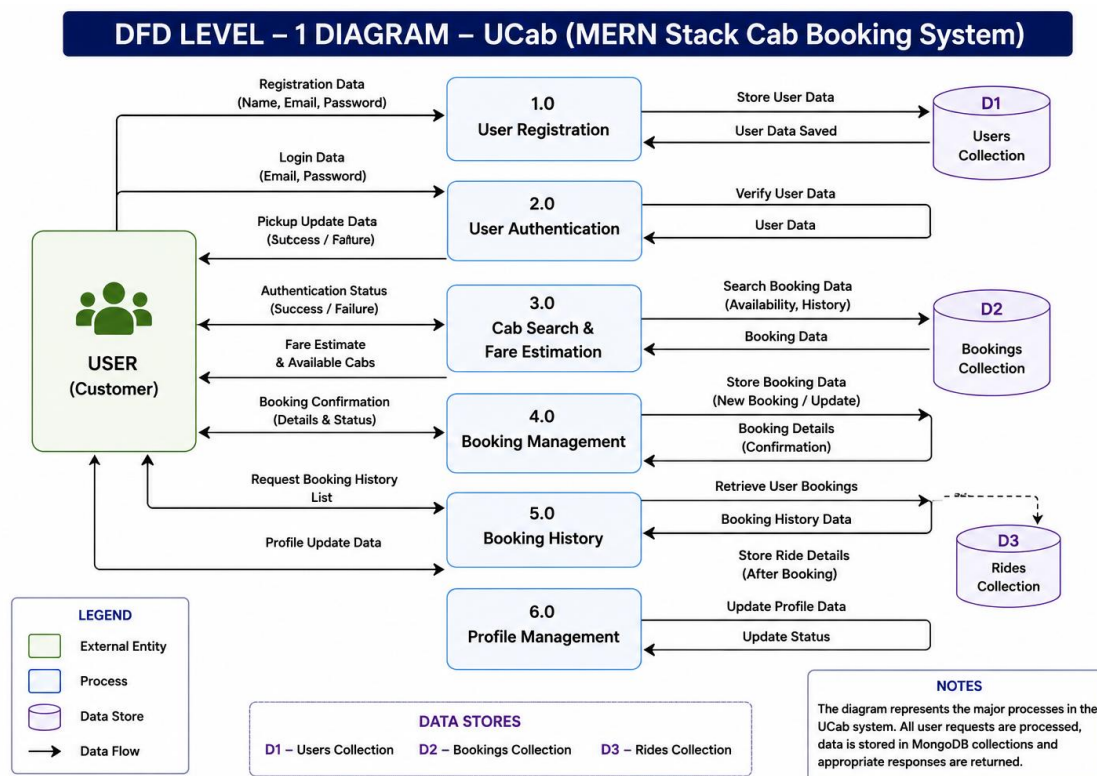
**Project Name:** UCab – MERN Stack Cab Booking System

### Data Flow Diagram (DFD)

#### Level-0 Data Flow Diagram (Context Diagram)



## Level-1 Data Flow Diagram



## Data Flow Description

The UCab system allows users to register and log into the application securely using JWT authentication. After successful login, users can book a cab by selecting pickup and destination locations. The application estimates the fare and stores booking information in the MongoDB database. Users can also access their booking history and manage their profile through the dashboard.

## User Stories

| User Type           | Functional Requirement (Epic) | User Story No. | User Story / Task                                             | Acceptance Criteria                   | Priority | Release  |
|---------------------|-------------------------------|----------------|---------------------------------------------------------------|---------------------------------------|----------|----------|
| Customer (Web User) | Registration                  | USN-1          | As a user, I can register using my name, email, and password. | User account is successfully created. | High     | Sprint-1 |
| Customer (Web User) | Login                         | USN-2          | As a user, I can log in using my                              | User is redirected to the dashboard   | High     | Sprint-1 |

| User Type           | Functional Requirement (Epic) | User Story No. | User Story / Task                                                       | Acceptance Criteria                                                          | Priority | Release  |
|---------------------|-------------------------------|----------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|----------|----------|
| Customer (Web User) | Authentication                | USN-3          | registered email and password.                                          | after successful login.                                                      | High     | Sprint-1 |
|                     |                               |                | As a user, I want my account to be securely authenticated.              | JWT token is generated and protected routes are accessible only after login. |          |          |
| Customer (Web User) | Book Cab                      | USN-4          | As a user, I can select pickup and destination locations to book a cab. | Booking request is successfully submitted.                                   | High     | Sprint-1 |
| Customer (Web User) | Fare Estimation               | USN-5          | As a user, I can view the estimated fare before confirming my booking.  | Estimated fare is displayed correctly.                                       | High     | Sprint-1 |
| Customer (Web User) | Booking Confirmation          | USN-6          | As a user, I can confirm my cab booking.                                | Booking details are saved in the database.                                   | High     | Sprint-1 |
| Customer (Web User) | Booking History               | USN-7          | As a user, I can view all my previous bookings.                         | Booking history is displayed correctly.                                      | Medium   | Sprint-2 |
| Customer (Web User) | Profile Management            | USN-8          | As a user, I can view and update my profile information.                | Updated profile information is stored successfully.                          | Medium   | Sprint-2 |
| Customer (Web User) | Logout                        | USN-9          | As a user, I can securely log out of my account.                        | Session is terminated and user is                                            | High     | Sprint-1 |

| User Type     | Functional Requirement (Epic) | User Story No. | User Story / Task                                          | Acceptance Criteria                                     | Priority | Release  |
|---------------|-------------------------------|----------------|------------------------------------------------------------|---------------------------------------------------------|----------|----------|
|               |                               |                |                                                            | redirected to the login page.                           |          |          |
| Administrator | User Management               | USN-10         | As an administrator, I can view registered users.          | List of users is displayed successfully.                | Low      | Sprint-3 |
| Administrator | Booking Management            | USN-11         | As an administrator, I can monitor all bookings.           | All booking records are displayed.                      | Low      | Sprint-3 |
| Administrator | System Monitoring             | USN-12         | As an administrator, I can manage the overall application. | Administrator has access to system management features. | Low      | Sprint-3 |

## Actors in the System

### Primary Actor

- Customer (Web User)

### Secondary Actors

- Administrator (Future Enhancement)

### Data Stores

- **D1:** Users Collection
- **D2:** Bookings Collection
- **D3:** Rides Collection

### Processes

- **P1:** User Registration
- **P2:** User Login

- **P3:** Authentication
- **P4:** Cab Booking
- **P5:** Fare Estimation
- **P6:** Booking Management
- **P7:** Ride History
- **P8:** Profile Management